Introduction to machine learning

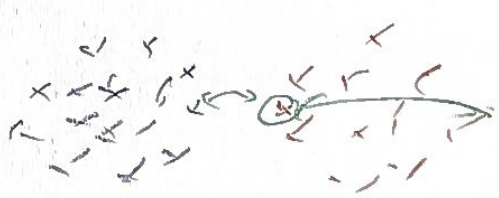
K-means clustering

Unsupervised learning

- Categories, but no measure
- Structure, but not observed (latent)

Idea: divide data into categories, based on a algorithm

Clustering: collect data points into clusters, where data in the same cluster is more similar to itself than to the rest



- K means clustering: given K centers $\underline{m}_k$ the points are identified by the center to which they are the closest

Clusters: $S_i = \{ \underline{x}_p : \|\underline{x}_p - \underline{m}_i\|^2 \leq \|\underline{x}_p - \underline{m}_j\|^2, \forall j \in \{1..k\} \}$

↳ Voronoi diagram



We want to minimize the average distance to the centers.

$$\min_{\{\underline{m}_i\}} J = \min_{\{\underline{m}_i\}} \sum_{i=1}^{k} \sum_{j \in S_i} \|x^{(j)} - m_i\|^2$$

This is an NP-hard problem!

Coordinate descent

Initialization: random assignment of $\underline{m}_i$

- for high dimensions or sparse data

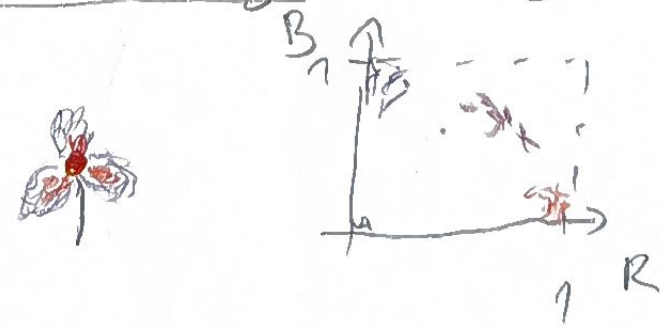
$m_i^{(0)} = x^{(j)}$ where j is chosen randomly to avoid empty S_i

Assignment step: determine sets S_i

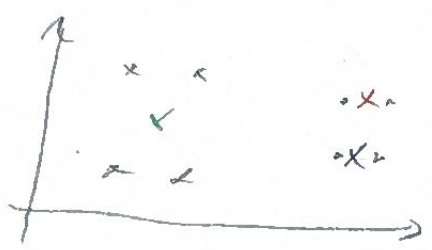
Update step: $\underline{m}_i^{(t+1)} = \frac{1}{|S_i^{(t)}|} \sum_{x_j \in S_i^{(t)}} \underline{x}_j$

center at the next timestep is the mean of the last one

Main usage: image segmentation in RGB space



Local minima



Soft K-means clustering

- Clusters are often not strict
- one datapoint may belong to multiple clusters

Let us define the distance function $d(\cdot,\cdot)$ which can be arbitrary. e.g. $d(x,y) = \|x-y\|$

Assignment:

$$r_k^{(n)} = \frac{\exp\left(-\beta\, d(\underline{m}_k, \underline{x}^{(n)})\right)}{\sum_j \exp\left(-\beta\, d(\underline{m}_j, \underline{x}^{(n)})\right)} = \underset{k=1\ldots K}{\text{softmax}}\left(-\beta\, d(\underline{m}_k, \underline{x}^{(n)})\right)$$

β : hyperparameter

Update:

$$\underline{m}_k = \frac{\sum_n r_k^{(n)}\, \underline{x}^{(n)}}{\sum_n r_k^{(n)}}$$

Summary

+ fast, easy
+ works with any data
- distance sensitive
- clusters found to set of the same weight



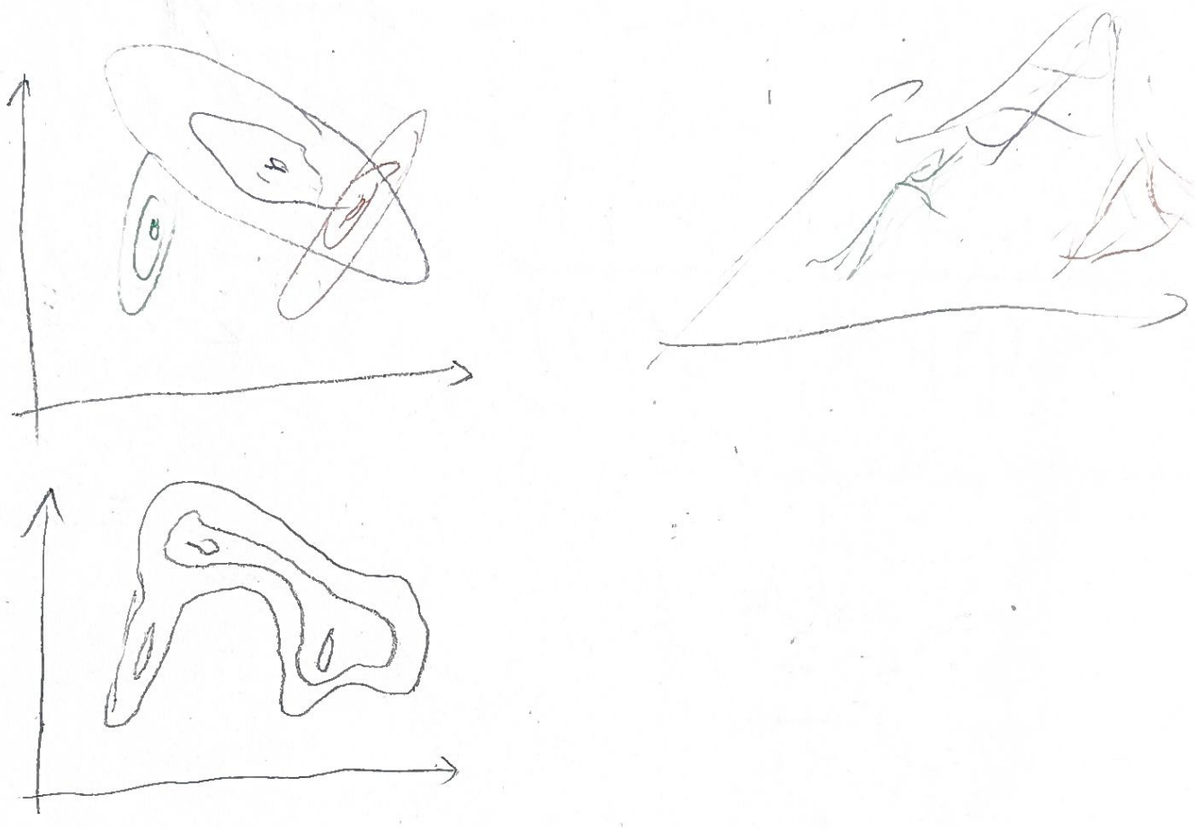
(a) → does not work

Gaussian Mixture Modeling

Gaussian distribution : $N(\mu_k, \Sigma_k)$

π_k : strength of Gaussian k

$$p(x) = \sum_{k=1}^{K} \pi_k N(x, \mu_k, \Sigma_k)$$



Parameters to fit: π_k, μ_k, Σ_k

Expectation maximization with maximum likelihood

- Unfortunately log likelihood does not help us for $k > 1$

However if we know the assignments $z^{(n)}$ (point (n) belongs to cluster $z^{(n)}$)

$$\sum_{n=1}^{N} \log p(\underline{x}^{(n)}, z^{(n)}) = \sum_{n=1}^{N} \left[\log(p(x^{(n)} | z^{(n)})) + \log(p(z^{(n)})) \right]$$

$$= \sum_{n=1}^{N} \left[\log \left(N(x^{(n)}, \mu_{z^{(n)}}, \Sigma_{z^{(n)}}) \right) + \log \left(\Pi_{z^{(n)}} \right) \right]$$

$$= \sum_{n=1}^{N} \sum_{k=1}^{K} \left[\mathbb{I}(z^{(n)} = k) \log \left(N(x^{(n)}, \mu_k, \Sigma_k) \right) + \right.$$
$$\left. + \mathbb{I}(z^{(n)} = k) \log \Pi_k \right]$$

From which we can get closed formula

We used $r_k^{(n)} = p(z^{(n)} = k | x^{(n)})$ for $\mathbb{I}(z^n = k)$

$$\Pi_k^* = \frac{1}{N} \sum_{n=1}^{N} r_k^{(n)}$$

$$\underline{\mu}_k^* = \frac{\sum_{n=1}^{N} r_k^{(n)} \underline{x}^{(n)}}{\sum_{n=1}^{N} r_k^{(n)}}$$

$$\underline{\underline{\Sigma}}_k^* = \frac{1}{\sum_{n=1}^{N} r_k^{(n)}} \sum_{n=1}^{N} r_k^{(n)} (\underline{x}^{(n)} - \underline{\mu}_k^*)(\underline{x}^{(n)} - \underline{\mu}_k^*)^T$$

($\underline{x}$ d dim, $\underline{\mu}$ d dim, $\underline{\underline{\Sigma}}$ d x d dim)

Unfortunately we do not know $r_k^{(n)}$, but we can calculate that for given $\pi_k, \underline{\mu}_k, \underline{\underline{\Sigma}}_k$ set.

However given $r_k^{(n)}$ we can optimize the parameters:

Initialization: random $\pi_k, \underline{\mu}_k, \underline{\Sigma}_k$

E-step: calculate class membership given the parameters

$$r_k^{(n)} = P(z^{(n)} = k \mid x^n) = \frac{\pi_k N(\underline{x}^{(n)}, \mu_k \Sigma_k)}{\sum_{j=1}^{K} \pi_j N(\underline{x}^n, \mu_j \Sigma_j)}$$

M-step: update parameters, knowing the membership

$$\pi_k = \frac{1}{N} \sum_{n=1}^{N} r_k^{(n)}$$

$$\mu_k = \left(\sum_{n=1}^{N} r_k^{(n)} \right)^{-1} \sum_{n=1}^{N} r_k^{(n)} x^{(n)}$$

$$\Sigma_k = \left(\sum_{n=1}^{N} r_k^{(n)} \right)^{-1} \sum_{n=1}^{N} r_k^{(n)} (x_k^{(n)} - \mu_k)(x_k^{(n)} - \mu_k)$$